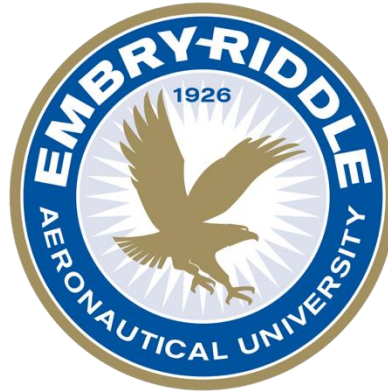
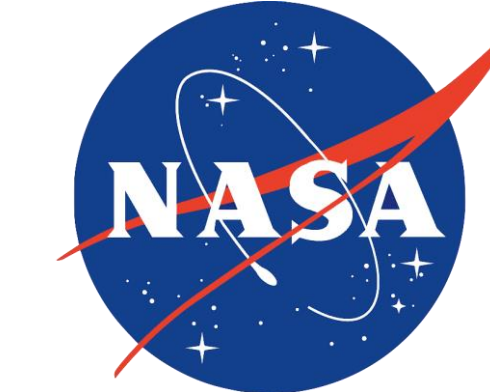




Characterizing the Solar Sources and Geospace Response of Isolated Stream Interaction Regions with Embedded Periodic Mesoscale Solar Wind Structures



Ainsley Helgerson^{1,2}, Samantha Wallace¹, Nicholeen M. Viall², Larry Kepko², Simone Di Matteo^{2,3}

¹Department of Physical Sciences Embry-Riddle Aeronautical University 1Aerospace Blvd. Daytona Beach, FL 32114, USA

²NASA Goddard Space Flight Center 8800 Greenbelt Rd Greenbelt, MD 20771, USA

³Physics Department, The Catholic University of America, Washington, DC 20064, USA

~ Overview ~

Stream Interaction Regions (SIRs) are compression regions from fast/slow solar wind interaction. They are less intense than CMEs but efficient drivers of recurrent geomagnetic activity. SIRs are embedded in a mesoscale-structured solar wind. One example of such mesoscale structures are periodic density structures (PDSs) which can be compressed within SIRs leading to large solar wind density variations potentially making SIRs more geoeffective.

Main Research Objective: Characterize the solar sources of the solar wind streams forming SIRs and the associated magnetospheric responses, using ADAPT-WSA coronal modeling, solar wind observations, and geomagnetic indices.

Keywords: *Periodic Density Structures, Stream Interaction Regions, Coronal Source Region*

~ Data ~

- SIR List from Katsavrias, Di Matteo et al. (2025)²
- One-Minute OMNI Data:
 - B, V, AE, AU, AL, Sym H, Alpha Proton Ratio, and Plasma Beta
 - Calculated total energy input into the magnetosphere via ε (a solar wind coupling function known as the Akasofu Parameter)
- One-Hour OMNI Data:
 - Magnetosphere parameters Kp (3-hour product), Dst
- Three Substorm Lists:
 - Newell, Ohtani, and Forsyth from SuperMAG
- ADAPT-WSA Model Output Files
- Source Region Parameters:
 - Expansion factor, Squashing factor (Q), B footpoint, Coronal hole distance, heliospheric current sheet distance

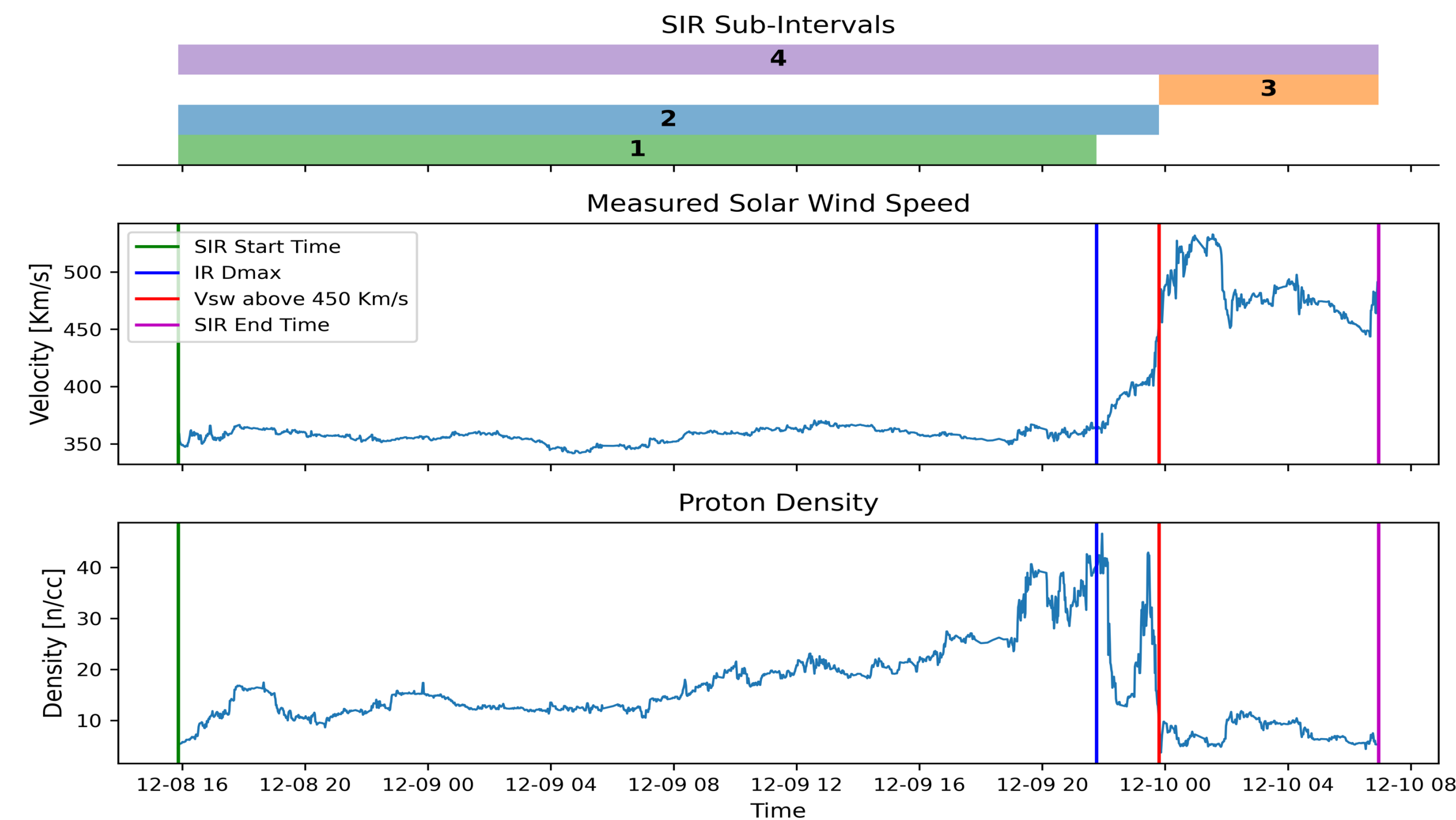
~ Methodology ~

We executed this analysis by following these steps:

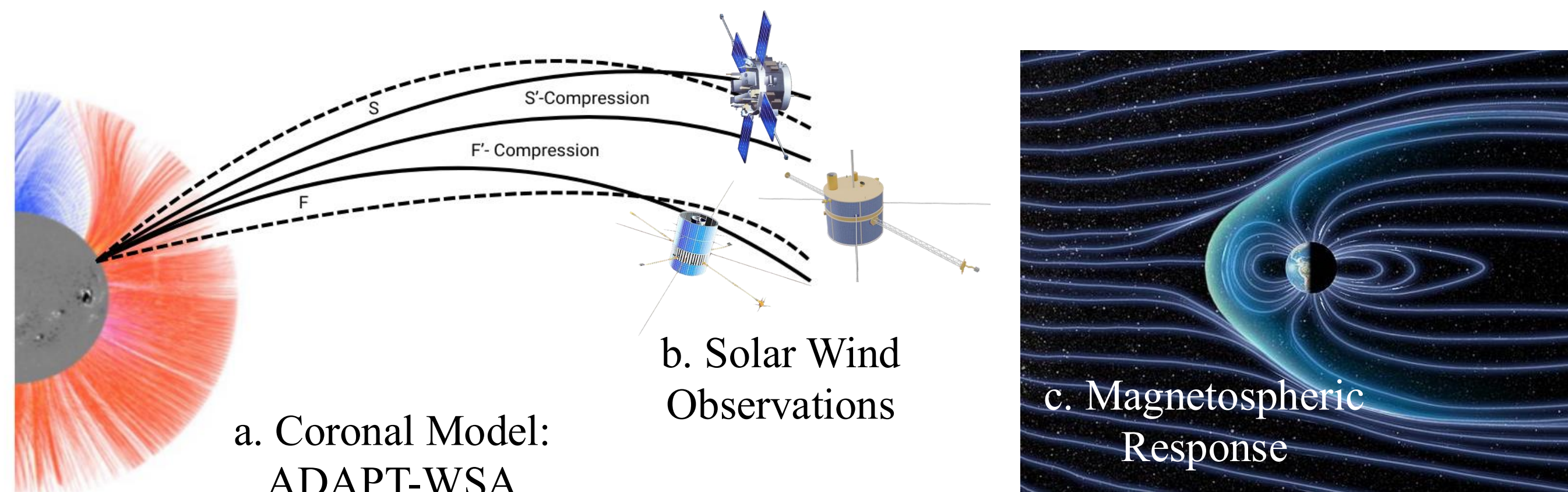
- Event list:** 208 total SIR events, 149 with derived source-region parameters via ADAPT-WSA.
- Calculated the Akasofu Parameter¹:**
 - Definition:** A solar wind coupling function that combines solar wind speed, magnetic field strength, and IMF orientation to estimate the rate of energy transferred into Earth's magnetosphere.
$$\varepsilon = \frac{4\pi}{\mu_0} V B^2 L_o^2 \sin^4 \left(\frac{\theta}{2} \right)$$
- Integrate over time to estimate the total energy transferred into the magnetosphere.

~ Methodology ~

- Divided Our List into a Four-Interval Framework:**



- Histograms of Coronal Model, Geospace, and Solar Wind Responses:**



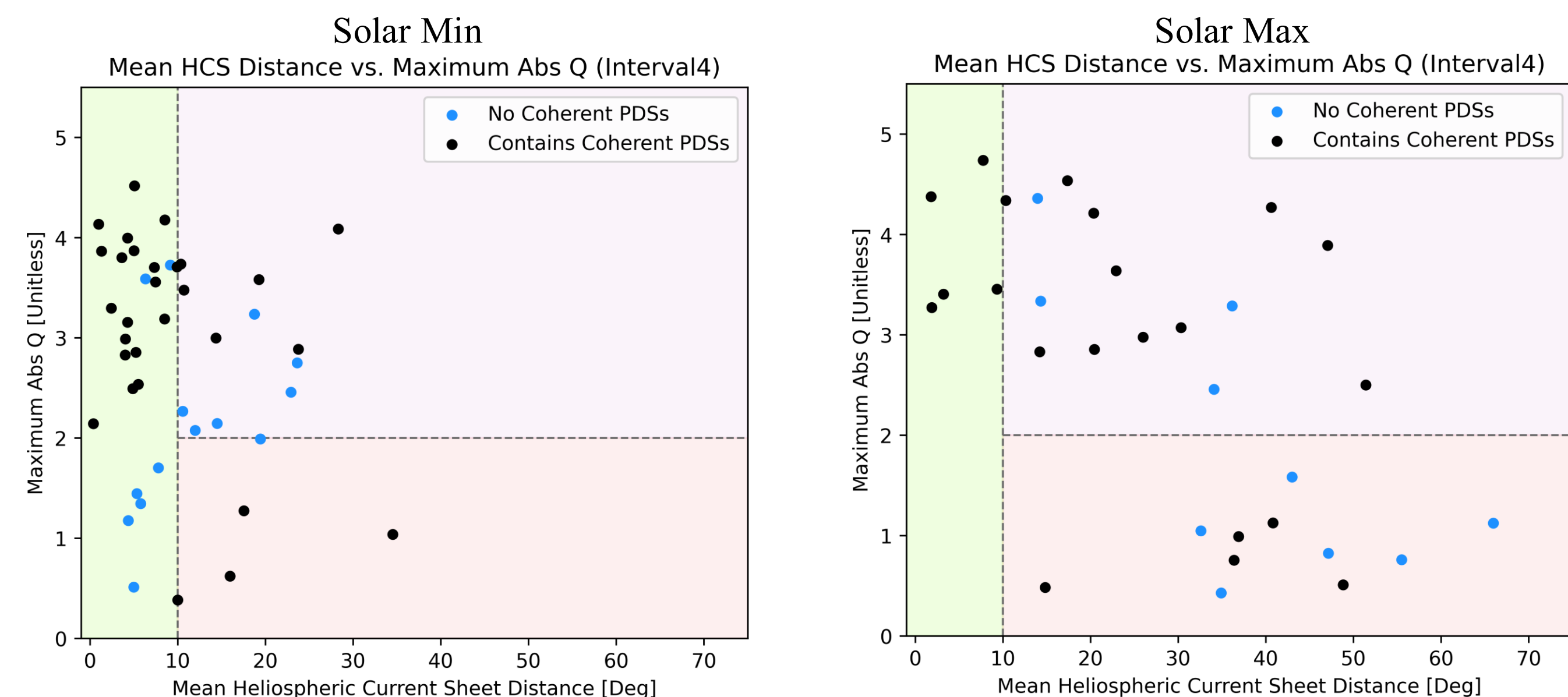
~ Conclusion & Future Work ~

Conclusion:

- Isolated SIRs with embedded PDSs drive moderate-to-strong geomagnetic activity, similar to canonical SIRs.
- Source-region modeling shows traces from S-web arcs and open/closed boundaries far from the heliospheric current sheet.

Future Work:

- Characterize PDS frequency distributions across SIR intervals back to their solar sources and compare coherent vs. non-coherent PDS populations (normalized to total events) across the scatter plot quadrants.
- Compare results with solar maximum and minimum times for cycles 23 and 24.

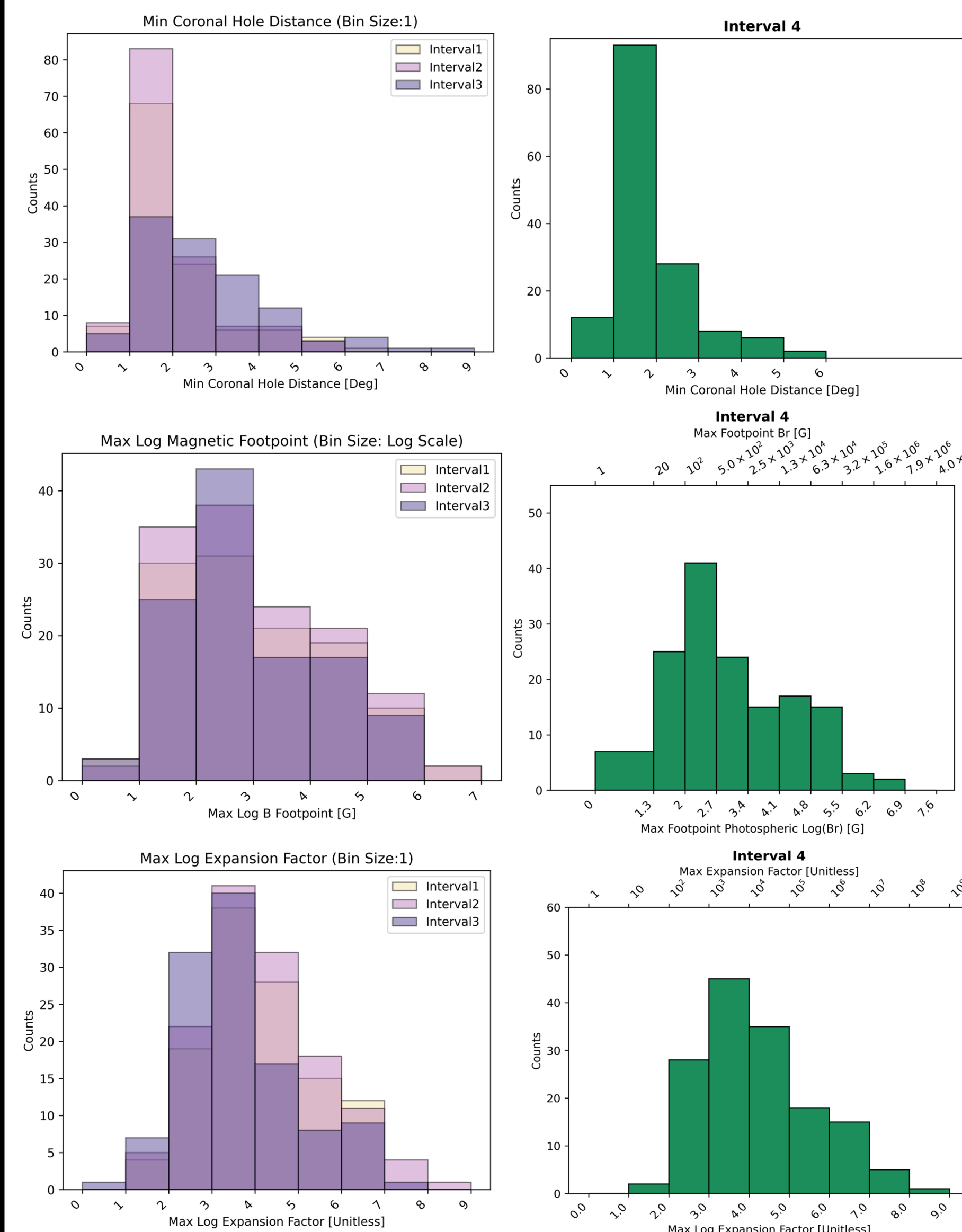


References:

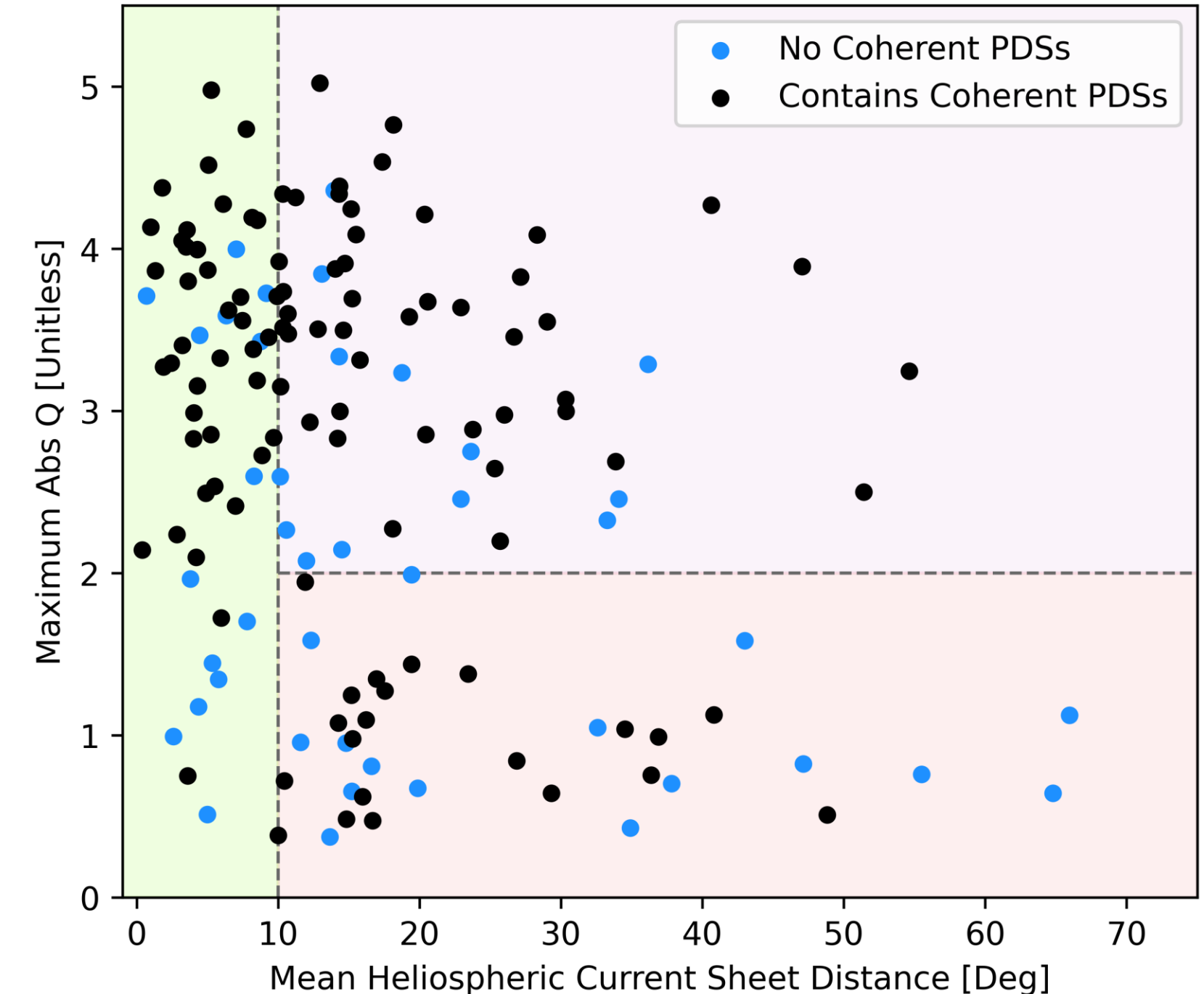
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~ Results ~

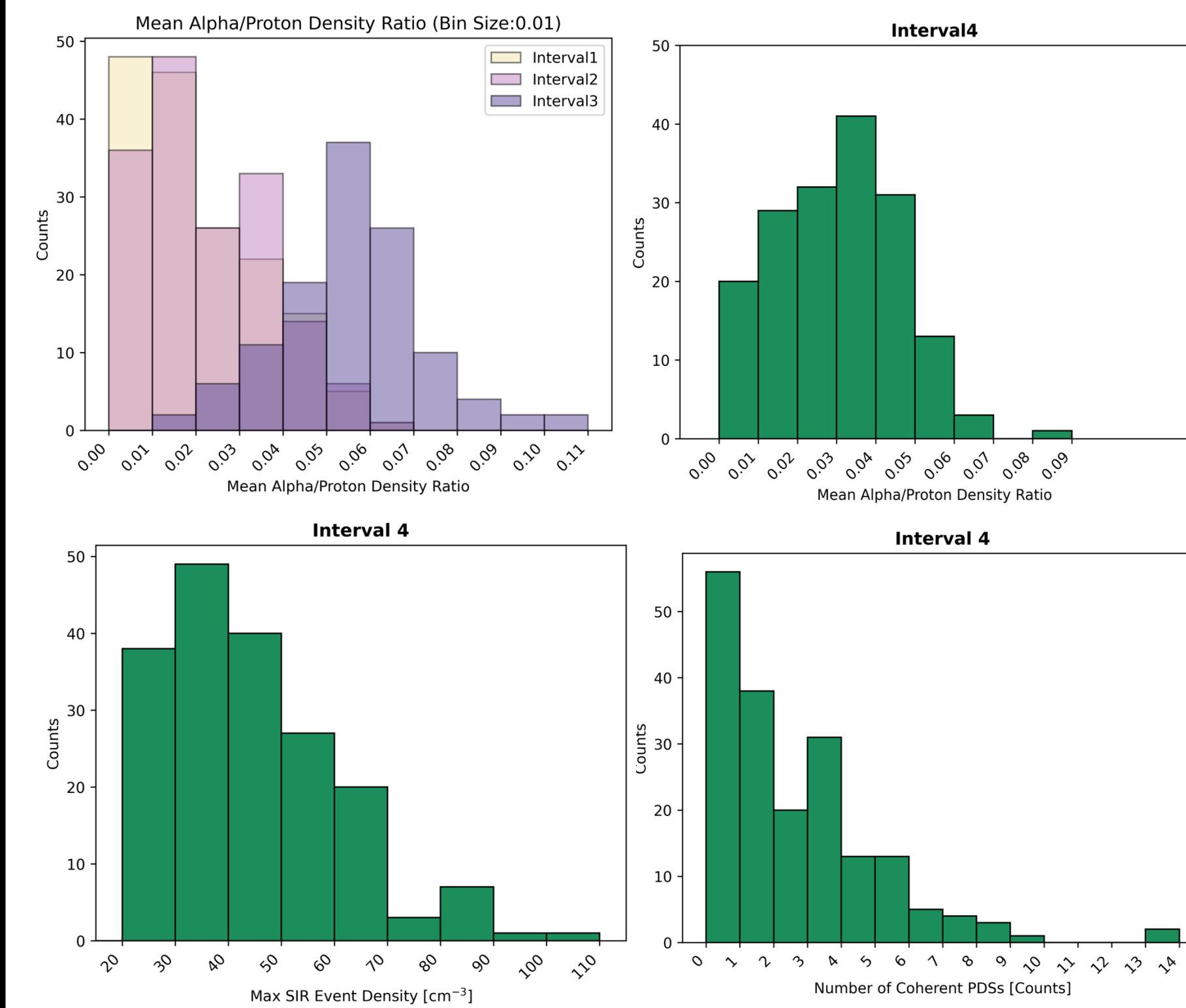
Source Origin Response



Mean HCS Distance vs. Maximum Abs Q (Interval 4)



Solar Wind Response



Geomagnetic Response

